



# Redux Notes (Roman Urdu - Beginner to Job Ready)

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## 1. Redux Kya Hai?

Redux aik **state management library** hai jo React apps mein data ko centrally manage karti hai.

### Simple meaning:

Puri app ka data aik central jagah (store) mein rakha jata hai

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## 2. Store Kya Hota Hai?

Store Redux ka **central data container** hota hai.

### Example:

```
store = {  
  counter: { value: 0 },  
  user: { name: "Hassan" },  
  cart: { items: [] }  
}
```

### Important:

- Store = poori app ki global state
  - Sirf 1 store hota hai
- 



## 3. Slice Kya Hota Hai?

Slice = store ka chhota part

### Example:

- counter slice

- user slice
- cart slice

Store = slices ka combination

## 4. Reducer Kya Hota Hai?

Reducer aik function hota hai jo state ko update karta hai.

### Example:

```
increment: (state) => {  
  state.value += 1;  
}
```

### Important:

- Reducer old state + action leta hai
- new state return karta hai

## 5. Action Kya Hota Hai?

Action aik simple JavaScript object hota hai.

### Example:

```
{  
  type: "counter/increment"  
}
```

### Agar payload ho:

```
{  
  type: "cart/add",  
  payload: { id: 1 }  
}
```



## 6. Action Creator Kya Hota Hai?

Action creator aik function hota hai jo action object return karta hai.

**Example:**

```
function increment() {  
  return {  
    type: "counter/increment"  
  };  
}
```



## 7. createSlice Kya Karta Hai?

createSlice automatically banata hai:

✓ Reducer ✓ Action creators

**Example:**

```
const counterSlice = createSlice({  
  name: "counter",  
  initialState: { value: 0 },  
  reducers: {  
    increment: (state) => {  
      state.value += 1;  
    }  
  }  
});
```

**Internally:**

```
increment() → action creator  
counter/increment → action type
```



## 8. useDispatch Kya Hai?

useDispatch action ko store tak bhejta hai.

### Example:

```
const dispatch = useDispatch();  
  
dispatch(increment());
```

### Flow:

action create hota hai → dispatch → store

## 9. useSelector Kya Hai?

useSelector store se data read karta hai.

### Example:

```
const count = useSelector((state) => state.counter.value);
```

## Important:

- useSelector = read data
- useDispatch = update data

## 10. Complete Redux Flow

```
Button Click  
↓  
dispatch(increment())  
↓  
Action Object Create  
↓  
Store  
↓  
Reducer Run  
↓  
State Update
```

↓  
useSelector re-render UI

## ? 11. Tumhare Important Questions + Answers

### ? Q1: Kya actions automatically banti hain?

✓ Haan (createSlice automatically banata hai action creators)

### ? Q2: Kya useDispatch action banata hai?

✗ Nahi ✓ Sirf action ko store tak bhejta hai

### ? Q3: Kya reducer mein action dena zaroori hai?

✗ Nahi (agar use nahi karna to skip kar sakte ho)

```
(state) => {}
```

ya

```
(state, action) => {}
```

### ? Q4: Store ke andar multiple stores hotay hain?

✗ Nahi ✓ Sirf 1 store hota hai ✓ Uske andar multiple slices hoti hain

### ? Q5: useSelector direct slice kyun nahi access karta?

✓ Kyunki React ko poora store milta hai ✓ Hum specify karte hain kya chahiye

```
(state) => state.counter.value
```

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## ? Q6: Agar 2 reducers ka same naam ho?

✓ Redux distinguish karta hai using:

```
sliceName/reducerName
```

Example:

- counter/increment
- cart/increment

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## ? Q7: createSlice ke waqt kya banta hai?

✓ Action creators ✓ Reducer ✗ Action object nahi

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## ? Q8: Action object kab banta hai?

✓ Jab action creator function call hota hai

```
increment()
```



## 12. Mini Code Example (Complete)

```
const counterSlice = createSlice({
  name: "counter",
  initialState: { value: 0 },
  reducers: {
    increment: (state) => {
      state.value += 1;
    },
    addByValue: (state, action) => {
      state.value += action.payload;
    }
  }
});
```

```
export const { increment, addByValue } = counterSlice.actions;  
export default counterSlice.reducer;
```

## Component Example

```
function Counter() {  
  const count = useSelector((state) => state.counter.value);  
  const dispatch = useDispatch();  
  
  return (  
    <div>  
      <h1>{count}</h1>  
      <button onClick={() => dispatch(increment())}>  
        Increase  
      </button>  
    </div>  
  );  
}
```

## Final Summary

Redux = centralized state system  
Store = global state container  
Slice = state ka part  
Reducer = state update karta hai  
Action = instruction object  
Dispatch = action bhejta hai  
useSelector = data read karta hai

## One Line Memory Trick

Dispatch → bhejta hai  
Reducer → update karta hai  
Selector → read karta hai 😊